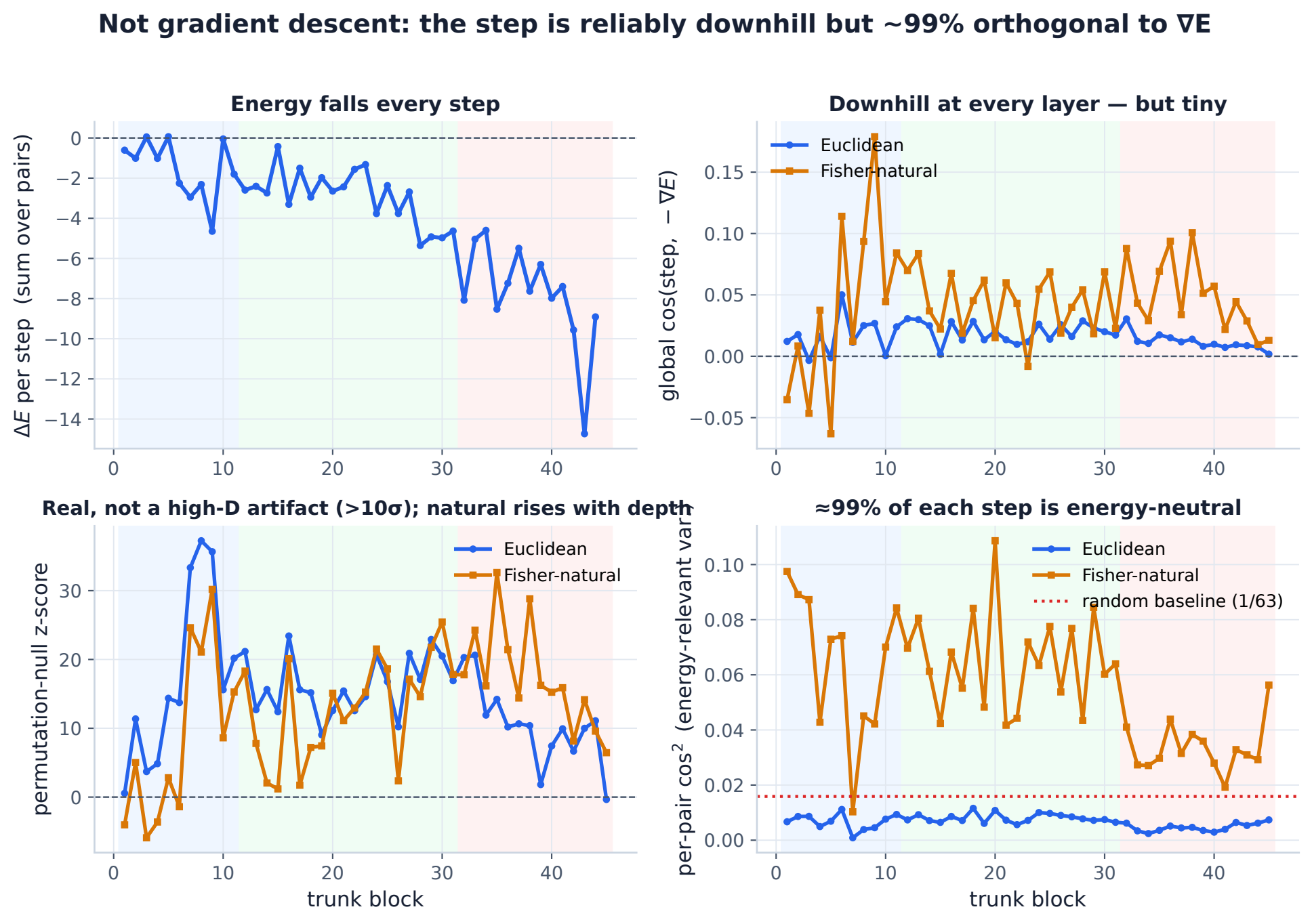
Not gradient descent: the step is reliably downhill but ~99% orthogonal to ∇E
Energy falls every step
ΔE per step (sum over pairs)
Downhill at every layer — but tiny
global cos(step, −∇E)
Euclidean
Fisher-natural
Real, not a high-D artifact (>10σ); natural rises with depth
permutation-null z-score
trunk block
≈99% of each step is energy-neutral
per-pair cos² (energy-relevant var
random baseline (1/63)
trunk block